

Benzene (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to discuss the structure of benzene based on physical and chemical evidence.

Early in the 20th century, benzene was a popular molecule that was commonly used in everyday substances and processes, from decaffeinating coffee to providing aftershave with its aroma. In modern times, many of those uses have stopped as benzene has since been found to be a carcinogen (cancer-causing) to humans. However, benzene and its derivatives are still one of the top 20 most produced chemicals by volume in the USA. Worldwide, 71 million metric tons of benzene were produced in 2022. What is benzene and what is it about its structure that allows it to have so many uses?

International Mindedness

Benzene is widely used as an industrial solvent in the developed world and has many uses. The fact that it has so many uses is in juxtaposition with the fact that it is actually a toxic compound and a known carcinogen.

Discovering the harmful nature of benzene required the collaboration and sharing of data internationally as workers across the globe began to get sick.

Since the discovery, regulatory bodies (organisations set up by governments to set national standards) around the world have now set strict limits on what is an acceptable level of benzene exposure.

Proposed structures of benzene

Benzene is a compound with the chemical formula C_6H_6 .

Section S2.2.7 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/>) described how carbon atoms always need to form four bonds to satisfy the octet rule. As the ratio of C:H in benzene is 1:1 for carbon to still form the four bonds needed, it must form some double or triple bonds.

Scientists throughout history have proposed many different structures for the benzene molecule, some of which can be seen in **Figure 1**.



Figure 1. Historic benzene formulas, reading left to right: Claus (1867), Dewar (1867), Ladenburg (1869), Armstrong (1887), Thiele (1899).

 More information for figure 1

The image displays five historic formulas of benzene. Each structure is represented as a hexagon: 1. The first structure has two diagonal lines crossing through the hexagon. 2. The second structure includes parallel lines on two opposite sides. 3. The third structure features a Y-shape emanating from three corners toward the center. 4. The fourth structure has lines radiating from the center to each corner, resembling a star inside the hexagon. 5. The last structure depicts a dashed circle inside the hexagon.

[Generated by AI]

In 1865 a German scientist, August Kekulé, proposed a new structure of benzene inspired by a dream where he saw a snake eating its own tail. Kekulé suggested a cyclic arrangement of alternating single and double bonds (**Figure 2**). This structure with three double bonds was able to explain some of the experimentally observed properties of benzene.

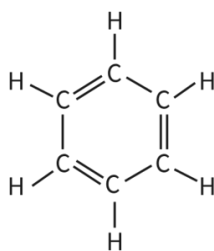


Figure 2. Kekulé's proposed structure of the benzene molecule C_6H_6 .

 More information for figure 2

The image is a diagram depicting Kekulé's proposed structure for the benzene molecule, C_6H_6 . It shows a hexagonal ring with alternating single and double bonds. There are six carbon (C) atoms forming the vertices of the hexagon, each bonded to one hydrogen (H) atom. The bonds alternate between single and double connections, forming a symmetrical cyclic structure. This diagram represents the classic Kekulé benzene structure, illustrating a possible explanation for the chemical characteristics of the molecule.

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Apply the information about the length of single, double and triple bonds from section S2.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>), and use section 11 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-or-average-covalent-bond-lengths-id-45113/>) of the DP chemistry data booklet to suggest the expected values for the C–C and C=C bond lengths.

On the basis of the knowledge that double bonds are shorter than single bonds, you may have suggested a lower value for the C=C bond than the C–C bond. However, when the bond length was analysed, it was found that all bond lengths between the carbon atoms were actually the same. This is not what would be expected for a molecule with alternating single and double carbon–carbon bonds.

Nature of Science

Aspect: Measurement

Science is an ever-changing field that must make use of all the evidence available, and the combination of both theoretical and experimental information is vital. Scientists must be prepared to adapt and discount theories if the experimental evidence no longer supports them.

Whereas Kekulé's proposed structure was helpful, it was clear an even more evolved structure was needed. In 1931, Linus Pauling suggested that benzene had a resonance hybrid structure with a ring of delocalised electrons, rather than the alternating single and double bonds that Kekulé had proposed. The next section will outline how existing experimental evidence contradicts the model suggested by Kekulé and supports Pauling's model.

Benzene does not behave like a molecule with single and double bonds

1. **Physical evidence:** All bond lengths and angles are the same.

If Kekulé's proposed structure was correct, then it would be expected that different bond lengths would be present to reflect the longer single bonds and shorter double bonds. However, the observed bond lengths are actually all the same and of lengths between that expected for a single and a double carbon-carbon bond (**Table 1**).

Table 1. Observed bond lengths.

Bond	Bond lengths (pm)
C—C	154
C=C	133
Carbon bond lengths for benzene	139

Also, all the bond angles are the same at 120° , as would be expected in a planar hexagonal ring where all bonds are identical.

The electron density map in **Figure 3** shows this. The blue lines join the parts of the molecule that have equal electron density.

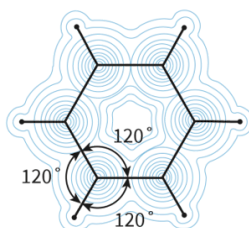


Figure 3. Electron density map of benzene.

 More information for figure 3

The image is an electron density map of benzene. It shows a hexagon representing the benzene molecule, with lines originating from each vertex to maintain an equal angle of 120 degrees, indicating the symmetrical shape. Surrounding the hexagon are concentric blue circles which represent areas of equal electron density, illustrating how electron density varies around the benzene ring. The regularity of the circles indicates consistent electron density throughout the molecule.

[Generated by AI]

2. Chemical evidence: Benzene is more stable than expected.

There is a particular reaction known as 'hydrogenation' that occurs when a carbon-carbon double bond in a molecule is broken and a H_2 molecule is added to the molecule (**Figure 4**). This reaction releases energy: approximately 120 kJ mol^{-1} for breaking **one** carbon-carbon double bond.

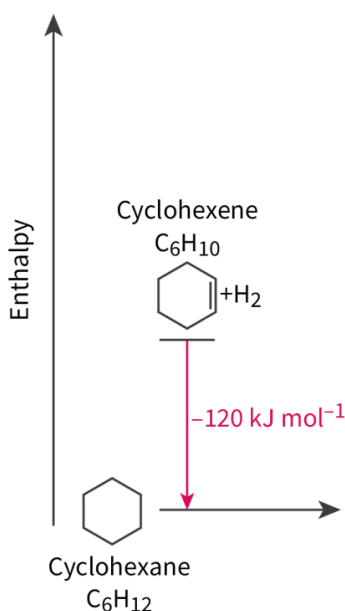


Figure 4. Hydrogenation reaction releases energy when one double bond is broken and one H_2 is absorbed.

 More information for figure 4

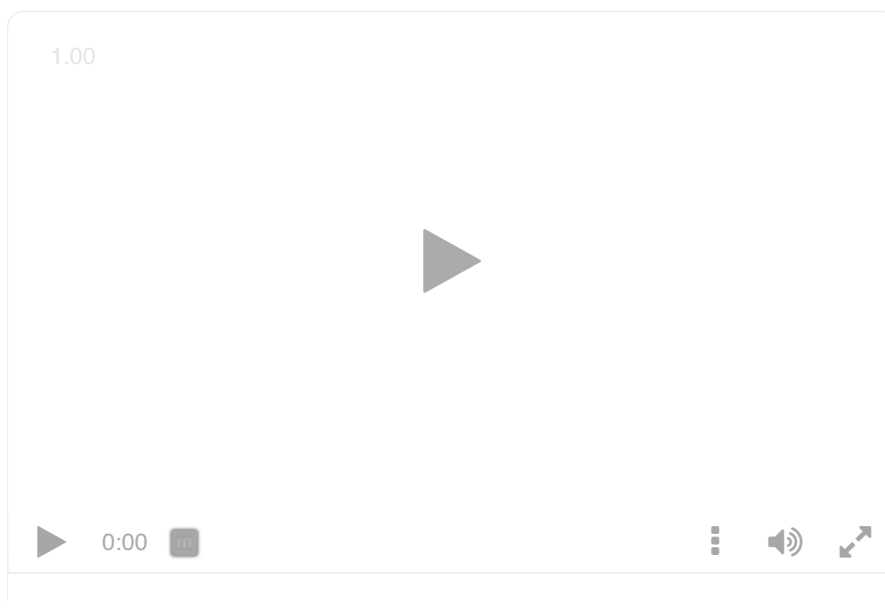
The diagram illustrates the hydrogenation reaction where cyclohexene (C_6H_{10}) is converted into cyclohexane (C_6H_{12}). The diagram has an X-axis and a Y-axis. The Y-axis represents enthalpy, and there's a negative enthalpy change of -120 kJ mol^{-1} depicted by a downward arrow. At the top, the chemical structure of cyclohexene along with H_2 is shown. Below, there's a chemical structure of cyclohexane, indicating the absorption of hydrogen and the release of energy in the process.

[Generated by AI]

If Kekulé was correct that three double bonds were present in a benzene molecule, all three bonds could undergo hydrogenation. These three double bonds would have to be broken and three H_2 molecules adsorbed onto the molecule. How much energy

do you think would be released for the hydrogenation of all three double bonds? Watch the animation to help you answer this.

It would be predicted that this should result in triple the energy released when one bond is hydrogenated: that is, that 360 kJ mol^{-1} would be released.



Interactive 1. Theoretical Energy Released for Hydrogenation of Three Double Bonds in Kekulé's Proposed Structure.

 More information for interactive 1

The video displays an energy diagram labeled “Kekulé’s proposed structure”, with energy on the y-axis and an unspecified variable on the x-axis. The diagram shows three stages of hydrogenation for benzene (C_6H_6) with intermediate compounds C_6H_8 and C_6H_{10} . Each step includes an energy release value in succession as follows. C_6H_{12} to $\text{C}_6\text{H}_{10} + \text{H}_2$ releases negative 120 kJ mol^{-1} . C_6H_{10} to $\text{C}_6\text{H}_8 + \text{H}_2$ releases negative 232 kJ mol^{-1} . C_6H_8 to $\text{C}_6\text{H}_6 + \text{H}_2$ releases negative 360 kJ mol^{-1} . Each structure is represented by a hexagonal ring with alternating double bonds for C_6H_6 , fewer double bonds for intermediates, and a fully saturated hexane structure at the base. The predicted energy released is highlighted in pink. Arrows indicate energy reduction at each step.

In reality, much less energy is being released; approximately 150 kJ mol^{-1} less. This is because the ring of delocalised electrons in benzene provides it with an additional stability (**Figure 5**). This value of 150 kJ mol^{-1} represents this additional stability and is known as the resonance energy or the stabilisation energy of benzene.

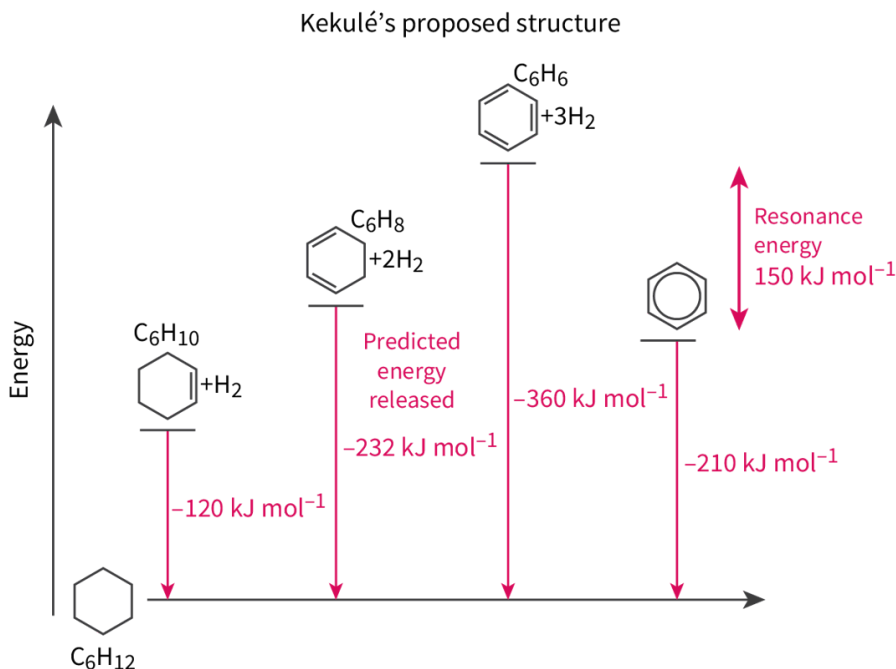


Figure 5. Benzene releases less energy than expected during a hydrogenation reaction.

More information for figure 5

This graph illustrates the energy levels associated with hydrogenation reactions of various hydrocarbons. The Y-axis represents energy, though it's unlabeled with specific values. The X-axis features the chemical formulas C_6H_{12} , C_6H_{10} , C_6H_8 , and C_6H_6 , each followed by the chemical reactions involving the addition of hydrogen (H_2).

- For C_6H_{12} , the energy change is noted as -120 kJ mol⁻¹ upon the addition of H_2 .
- For C_6H_{10} , the predicted energy released is -232 kJ mol⁻¹ upon reacting with H_2 .
- C_6H_8 's reaction with $2H_2$ shows an energy release of -360 kJ mol⁻¹.
- The structure associated with C_6H_6 reacting with $3H_2$ reveals an energy change of -210 kJ mol⁻¹.

Importantly, the graph highlights a resonance energy of 150 kJ mol⁻¹, which accounts for the additional stability due to the delocalization of electrons in benzene (C_6H_6). This is depicted by an arrow indicating the resonance energy separate from the main energy changes shown for each molecule. The energy values are marked with arrows pointing downward to signify energy release, except for the resonance energy arrow, which points upward.

[Generated by AI]

Note that the enthalpy of hydrogenation value for C_6H_8 (cyclohexa-1,3-diene) doesn't quite fit the trend shown for C_6H_{10} and C_6H_6 . This is due to the delocalisation of the pi electrons in the two C=C bonds in C_6H_8 that lower the value by 8 kJ mol⁻¹.

Making connections

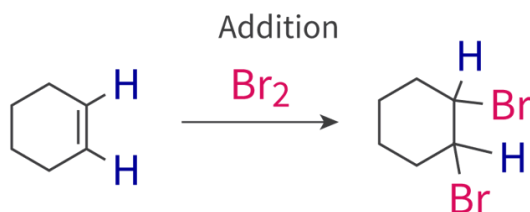
Consider all the information in [subtopics S2.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>) and [S2.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>), how does the resonance energy in benzene explain its relative unreactivity?

3. Chemical evidence: Benzene does not undergo the reactions expected for a molecule with double bonds.

The molecular structure suggested by Kekulé would be expected to undergo a type of reaction that involves breaking open its double bonds and adsorbing a separate small molecule onto the benzene molecule. This type of reaction is typical of molecules that have double bonds, and is known as an addition reaction (**Figure 6**).



Two Br have been added to the double bond

Figure 6. Addition reactions in a cyclic molecule.

 More information for figure 6

The diagram illustrates an addition reaction in a cyclic molecule. On the left, there is a hexagon representing a cyclic structure with double bonds, noted with two hydrogen (H) atoms attached. An arrow labeled "Addition" points to the right side, indicating the reaction process. To the right of the arrow, the resultant molecule shows the same hexagonal structure now attached with two bromine (Br) atoms instead of hydrogen, indicating that two Br atoms have been added to the double bond. The caption at the bottom reads, "Two Br have been added to the double bond," summarizing the reaction process.

[Generated by AI]

However, benzene does not do this. For benzene they are energetically unfavourable, meaning they would need too much energy to happen. This is because it would involve first disrupting the entire cloud of delocalised electrons which requires a lot of energy. The resonance energy described previously would have to be supplied and the product would be less stable without the delocalised ring of electrons.

Instead, benzene undergoes a type of reaction usually only associated with molecules with single bonds, which involves the replacement of atoms or groups of atoms with a part of the reactant molecule. This type of reaction is known as a substitution reaction and it is better able to preserve the stable ring structure (**Figure 7**).

Sections R3.4.11–12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-of-alkenes-hl-id-46326/>) and R3.4.13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-substitution-reactions-of-benzene-hl-id-46327/>) will cover more about these different types of reaction.

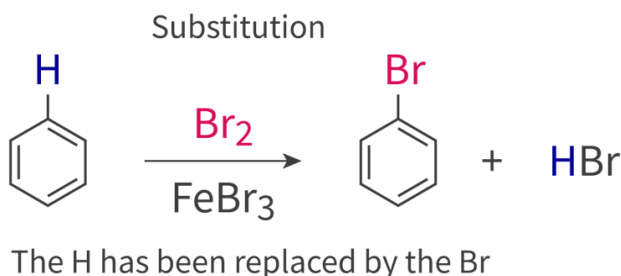


Figure 7. Substitution reactions in a cyclic molecule.

More information for figure 7

The image is a chemical diagram illustrating a substitution reaction in a benzene ring. On the left, a benzene ring with a hydrogen atom (H) marked in blue is shown. An arrow points to the right with the label 'Br₂' in pink above it and 'FeBr₃' below, indicating the introduction of bromine (Br₂) in the presence of iron(III) bromide as a catalyst. To the right, a similar benzene ring is shown, now with a bromine atom (Br) in pink replacing the hydrogen, indicating that substitution has occurred. Additionally, the by-product of the reaction is shown as HBr with the hydrogen in blue. Text below the diagram reads, "The H has been replaced by the Br," summarizing the result of the reaction.

[Generated by AI]

This can be physically seen when benzene is reacted with bromine solution. Bromine, Br₂, is orange in colour. Usually, for a molecule that has double bonds, the bromine will be adsorbed onto the molecule in an addition reaction, and so the solution loses its colour.

However, for benzene this does not happen.

When the bromine water is added to the colourless benzene liquid, the mixture remains orange; benzene is unable to decolourise the bromine water. The fact that it is unable to decolourise bromine is sometimes identified as being another piece of chemical evidence (**Figure 8**).

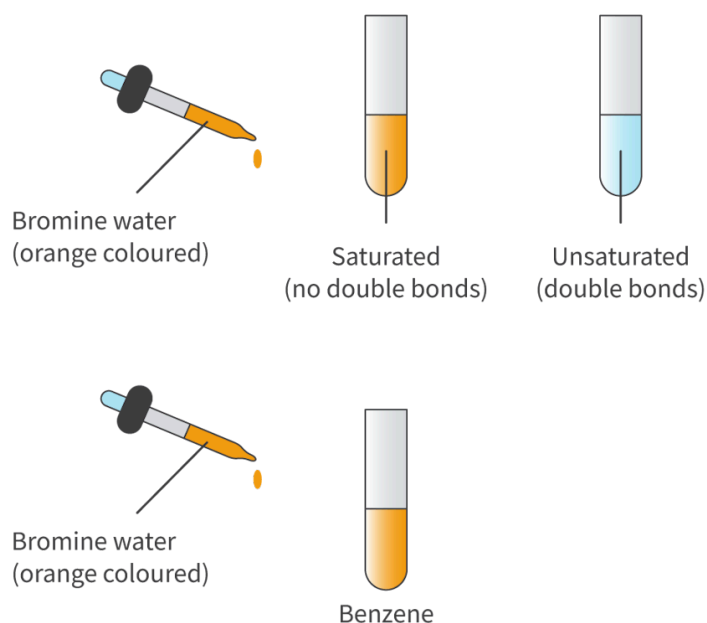


Figure 8. Benzene cannot decolourise bromine water.

 More information for figure 8

The image is a chemical diagram showing a comparison between the reactions of saturated, unsaturated compounds, and benzene with bromine water. The top row illustrates a dropper labeled "Bromine water (orange coloured)" with two test tubes. The first test tube is labeled "Saturated (no double bonds)" and contains an orange liquid, indicating no reaction, as the bromine color remains. The second test tube is labeled "Unsaturated (double bonds)" and shows the liquid as colorless, indicating a reaction where the bromine has been decolorized. The bottom row repeats the dropper with "Bromine water (orange coloured)" next to a test tube labeled "Benzene" containing an orange liquid. This shows that benzene does not react and decolorize the bromine water, similar to the saturated test tube in the top row.

[Generated by AI]

International Mindedness

Benzene is formed naturally from volcanoes but also synthetically in the laboratory. Traces of benzene have been found in pharmaceuticals, food, drinks and beauty products and, most recently, in the hand sanitiser we used during the Covid pandemic, prompting recalls on affected products.

What role do governing bodies play in preventing these traces, product testing and product recall, or does the responsibility lay with the companies producing the containers?

Summary of evidence

Each of these statements proves that benzene has a resonance hybrid structure rather than Kekulé's proposed structure:

1. The C—C bond lengths are all the same length, with the same bond angle.
2. Benzene is more stable than expected due to resonance energy (-150 kJ mol^{-1}).
3. Benzene is reluctant to undergo addition reactions and prefers substitution reactions.

Study skills

When asked to describe the evidence showing that benzene has a resonance hybrid structure, it is important that you can classify them as being either physical or chemical pieces of evidence. Physical evidence relates to measurements that would not affect the molecule, for example measuring the length of the bonds. To collect chemical evidence, you must change the molecule, such as reacting it with something to create a new molecule.

Benzene as a resonance structure

It was an Irish scientist, Kathleen Lonsdale, who conclusively proved the resonance structure of benzene using a technique known as X-ray crystallography. This could show the electron density around the carbon atoms, essentially providing a map of

electron density and from this, the distance between the carbon atoms could be determined. This supports the resonance hybrid structure of benzene containing a ring of delocalised electrons, rather than as the single–double–single–double ring structure Kekulé suggested (**Figure 9**).

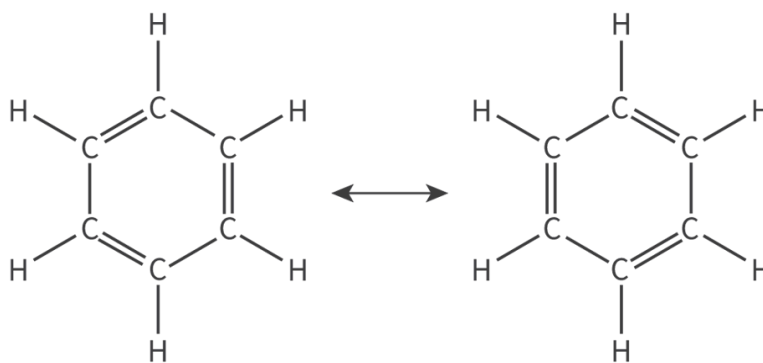


Figure 9. Resonance hybrid structure C_6H_6 .

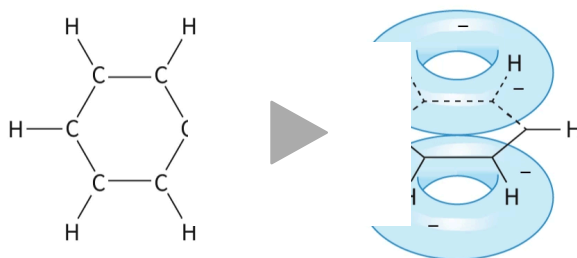
 More information for figure 9

The image is a diagram showing the resonance hybrid structure of benzene (C_6H_6). It includes two structures connected by a double-headed arrow, indicating resonance between them. Each structure is a hexagonal arrangement of carbon atoms, represented by the letter 'C', with one hydrogen atom, represented by the letter 'H', attached to each carbon. In one structure, alternating single and double bonds connect the carbons, starting with a double bond at the top. In the second structure, the pattern of single and double bonds is shifted one position around the ring. This visual representation highlights the concept of delocalized electrons, as the electrons aren't fixed between specific carbon atoms but can resonate between these structures.

[Generated by AI]

Let's examine what happens to the bonds within benzene that cause them all to be the same length. In total, the ring in benzene would have 18 electrons: 12 of these electrons make up the carbon–carbon single bonds and the remaining 6 electrons are all delocalised, as would be expected in a resonance structure (**Interactive 2**). These delocalised electrons are simultaneously attracted to the protons in the nucleus of the carbon atoms and also repelled by the 12 bonding electrons, and are found with some located above the plane of the molecule and some located below.

1.00



Interactive 2. Delocalised Bonding Electrons in the Benzene Molecule.

More information for interactive 2

An interactive video displays the delocalized electrons in a benzene molecule. On the left, it shows the chemical structure of the benzene, featuring six carbon atoms arranged in a hexagonal structure, bonded by single bonds. Each carbon atom is also single-bonded to one hydrogen atom. On the right, it displays a benzene structure in a hexagonal arrangement, where the carbon atoms are implied but not shown, and the hydrogen atoms are displayed. The lower half of the structure is represented with solid lines, while the upper half is shown with dashed lines. Two donut-shaped structures, representing delocalized pi-electron clouds, are present above and below the benzene ring. As the video progresses, three electrons are shown moving within each of the electron clouds, highlighting the concept of electron delocalization in the benzene molecule. This video helps users recognize how electron clouds are distributed above and below the planar ring structure and understand the symbolic representation of resonance.

The orbitals that hold these electrons end up merging as a result. This forms a sort of 'doughnut' or 'halo' shape, as seen in **Interactive 2**. This ring of delocalised electrons contains the bonding electrons being shared equally by all six carbon atoms.

Nature of Science

Aspect: Models

Visualising abstract concepts such as resonance hybrid structures can be difficult but the use of different types of models can help.

For example, this computer-generated representation of benzene can show the three-dimensional structure more effectively than a simple drawing. In this model, the carbon atoms are represented by the black atoms and the hydrogens are shown as white atoms. The clouds of delocalised electrons can be seen above and below the plane of the ring of carbon atoms as the blue and red areas.

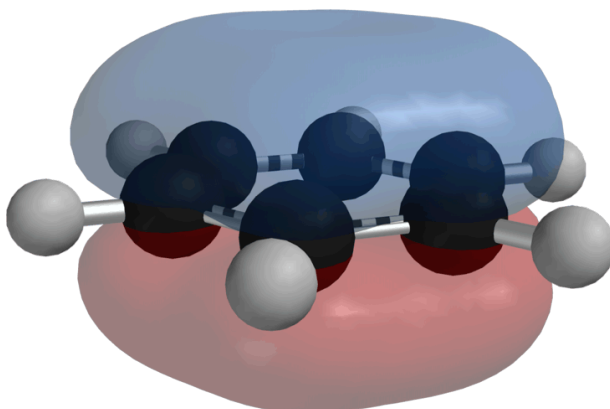


Figure 10. 3D representation of benzene.

Source: "Benzene-pi-MO-3D-balls" [↗](#)

(<https://en.wikipedia.org/wiki/File:Benzene-pi-MO-3D-balls.png>)

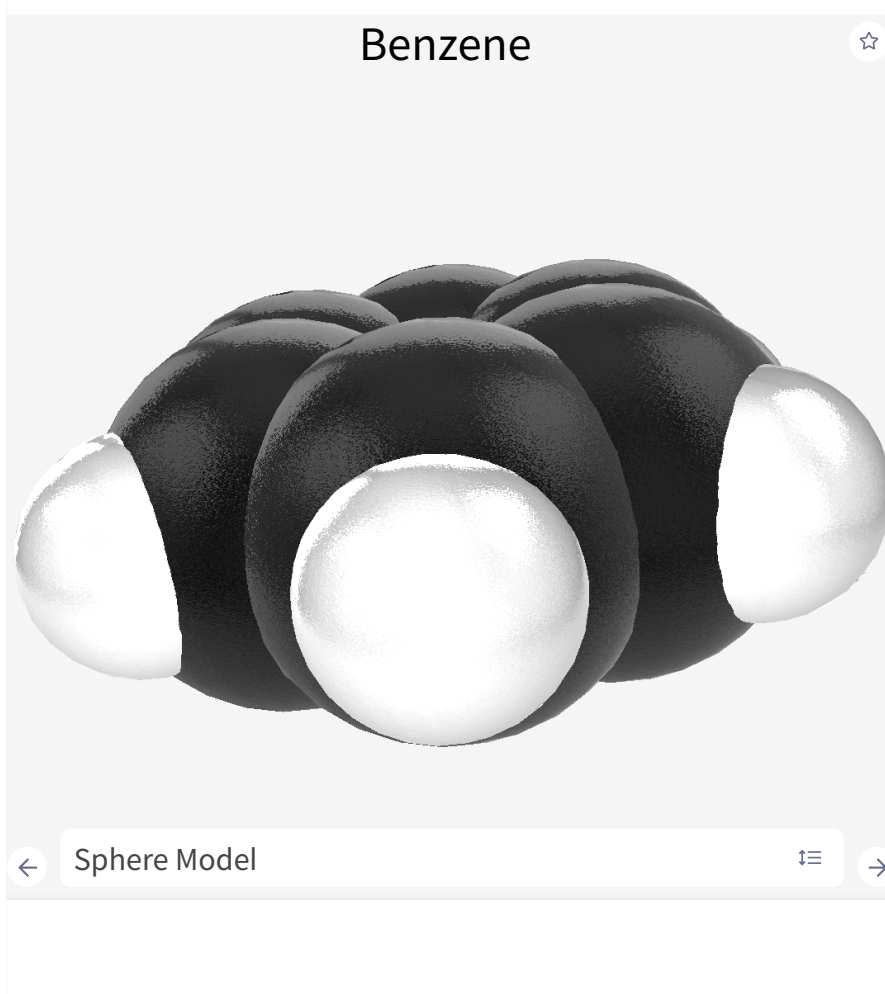
by Benjah-bmm27 is in the public domain

[🔍](#) More information for figure 10.3

The image is a 3D representation of a benzene molecule. It consists of a hexagonal ring showing six carbon atoms, represented as black spheres, and six hydrogen atoms, shown as white spheres. The model highlights the structure with bonds in a cyclic arrangement. Above and below the carbon ring are clouds depicting delocalised electrons, visualized as overlapping blue and red regions. These electron clouds indicate areas of electron density due to the pi bonds in benzene, enhancing the visual understanding of the molecule's electronic structure.

[Generated by AI]

To help you visualise the structure of the benzene molecule, you may find the model shown below useful.



Interactive 3. 3D Structure of the Benzene Molecule.

 More information for interactive 3

An interactive, three-dimensional illustration displays different representations of benzene. Users can rotate these representations in any direction. A carousel selector with arrows is at the bottom, allowing switching between a ball-and-stick model, a sphere model, and a structural formula.

In the ball-and-stick model, six large spheres representing carbon atoms are bonded in a hexagonal structure with alternating single and double bonds. Each carbon atom is connected to a smaller sphere representing hydrogen by a single bond. The bonds are represented as thin rods. In this model, the carbon and hydrogen atoms, pi electron system, and molecular bonds can be visualized by choosing the option.

In the sphere model, six large spheres representing carbon atoms are fused in a hexagonal pattern. Each carbon atom is fused with a smaller sphere representing a hydrogen atom on the outside. In this model, the carbon and hydrogen atoms can be visualized by choosing the options.

The structural formula features a hexagonal arrangement with a CH occupying each corner and alternating single and double bonds between them.

On the right side, the option for changing the background color and the option for zooming the interactive are given.

Below the selector are options labeled introduction, share, capture, augmented reality, and language.

The introduction option provides the following details:

Molecular formula: C_6H_6

Molar mass: 78.11 g/mol

Description: Benzene is an aromatic, colorless liquid with a sweet odor. It is highly toxic and carcinogenic. When burned, it produces large amounts of soot. Benzene vapors form an explosive mixture with air upon ignition.

Applications: Benzene is used as a solvent for many organic substances and is an important raw material in the production of dyes and pharmaceuticals.

These three-dimensional representations allow users to explore benzene structure, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.

Representing benzene

As you can see in **Figure 11**, there are many different ways to represent benzene's structure. Sometimes you may encounter the hybrid structure represented using the skeletal structure on the left, which includes a circle inside the hexagonal ring of carbons to represent the delocalised electrons.

It is important to understand that the hydrogens are still actually present, they are just not represented in the diagram for clarity's sake. When asked to draw benzene, you should always use the Lewis formulas: always show the full structure including the hydrogens.

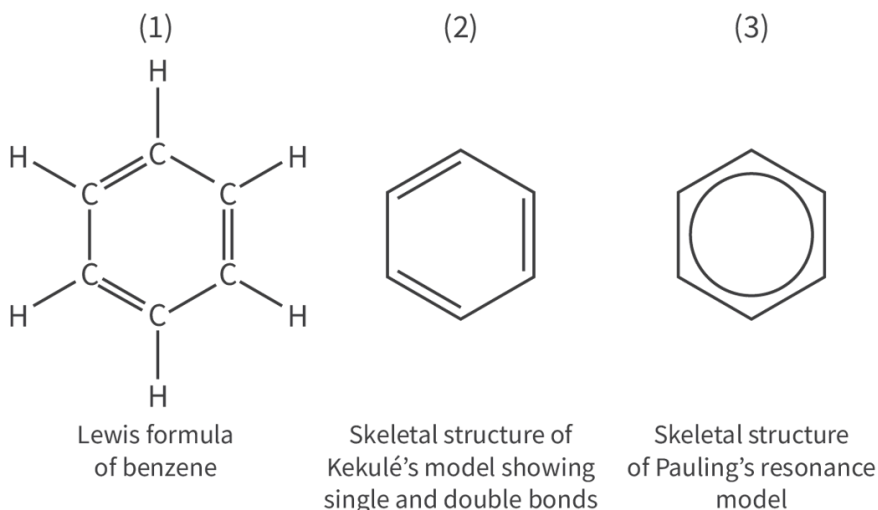


Figure 11. Alternative structures of benzene.

 More information for figure 11

The image contains three diagrams representing different structures of benzene. Each diagram is labeled.

1. The first diagram labeled '(1)' is the Lewis formula of benzene. It shows a hexagonal carbon ring, with alternating single and double bonds between the carbons, and a hydrogen atom attached to each carbon atom on the outside.
2. The second diagram labeled '(2)' represents the skeletal structure of Kekulé's model. It illustrates a hexagon with alternating single and double bonds within the carbon ring without indicating the hydrogen atoms.
3. The third diagram labeled '(3)' shows the skeletal structure of Pauling's resonance model. It displays a hexagon with a circle inside, representing the delocalized electrons within the benzene ring.

[Generated by AI]

This next activity will allow you to build this fascinating molecule and transfer the knowledge you have gained about benzene to a similar molecule.

Activity

- **IB learner profile attribute:**
 - Inquirer
- **Approaches to learning:** Thinking skills — Designing procedures and models; Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Group / Individual activity

This activity has two parts. Complete part 1 in a group with 3 other students and complete part 2 individually.

Part 1: Group of 4 students

In your group, you need to build two molecules using the materials listed for you below. The first molecule should represent Kekulé's proposed structure and the second should represent Pauling's hybrid resonance structure. Before you begin, consider what materials would best represent the features of each of these molecules.

Materials

- Either toothpicks or straws or pipe cleaners or string
- Either cotton balls or plasticine

Part 2: Individual

Pyridine, a molecule with a similar structure to benzene, is commonly used as a solvent in medicine, food flavourings and paint.

One Lewis formula of pyridine is shown below.

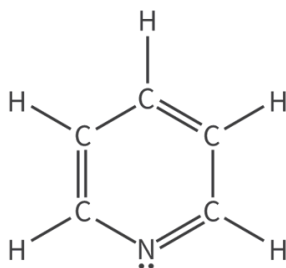


Figure 12. Lewis formula of pyridine.

 More information for figure 12

The image shows the Lewis structure of pyridine, a six-membered aromatic ring with five carbon (C) atoms and one nitrogen (N) atom. The nitrogen atom is at the top of the ring and features a pair of dots above it, representing a lone electron pair. Each carbon atom is connected to a

hydrogen (H) atom through a single line, indicating a single bond. The bonds alternate as double bonds and single bonds around the ring, characteristic of aromatic compounds. This cyclic structure forms a hexagonal shape. The structure illustrates the formula C_5H_5N , typical for pyridine.

[Generated by AI]

1. Draw any other potential Lewis formula of pyridine.
2. Predict the carbon—carbon and carbon—nitrogen bond length and strength, using [section 11](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-or-average-covalent-bond-lengths-id-45113/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-or-average-covalent-bond-lengths-id-45113/>) and [section 12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) in the DP chemistry data booklet. Then, research the value to see how close your estimate was.